VOLTAGE GAUGE (Continued)

about 19.0 volts, the gauge needle is moved to the 19 volt graduation on the far right end of the gauge scale and the check gauges indicator is illuminated. The gauge needle remains on the right end of the gauge scale and the check gauges indicator remains illuminated until the cluster receives a message from the PCM indicating the electrical system voltage is less than about 16.0 (but greater than about 9.5 volts), or until the ignition switch is turned to the Off position, whichever occurs first.

- **Communication Error** - If the cluster fails to receive a system voltage message, it will hold the gauge needle at the last indication about twelve seconds or until the ignition switch is turned to the Off position, whichever occurs first. After twelve seconds, it will move the gauge needle to the far left end of the gauge scale.
- **Actuator Test** - Each time the cluster is put through the actuator test, the gauge needle will be swept to several calibration points on the gauge scale in a prescribed sequence in order to confirm the functionality of the gauge and the cluster control circuitry.

The PCM continually monitors the system voltage to control the generator output. The PCM then sends the proper system voltage messages to the instrument cluster. For further diagnosis of the voltage gauge or the instrument cluster circuitry that controls the gauge, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the instrument cluster turns on the check gauges indicator due to a charge fail or voltage high condition, it may indicate that the charging system requires service. For proper diagnosis of the charging system, the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the voltage gauge, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.